

Smart skin for monitoring of marine animals

A team of researchers at King Abdullah University of Science and Technology (KAUST) in Saudi Arabia has developed Marine Skin—a smart patch that could be used to electronically tag marine animals. Marine Skin is derived from flexible silicone elastomers that can tolerate high pressures in deep waters.



An electronic tag that stretches and flexes while it records location and environmental data to monitor marine animals in their natural habitat (Credit: King Abdullah University of Science and Technology)

Former KAUST team member Joanna Nassar says that the patch tracks an animal's movement and diving behaviour, as well as the health of its surrounding marine environment, in real time. Marine Skin prototype supplements location data with temperature and salinity readings. Data can then be retrieved wirelessly when the patch is removed.

Nassar thinks that the prototype and planned enhancements will enable more comprehensive analysis of the marine ecosystem. Probing behavioural changes of organisms will help evaluate habitability in the face of rising global temperatures, increased pollution and the impact of overfishing.

SoFi, the robotic fish that can swim alongside real fish

Massachusetts Institute of Technology (MIT) researchers have designed a small robot fish with a wiggling tail that they



SoFi is a step towards developing an underwater observatory of sorts (Credit: Massachusetts Institute of Technology)

hope can mingle with real fish, offering marine biologists an insider's view of underwater life. According to graduate student Robert Katzschmann at Massachusetts Institute of Technology, traditional robots are not adept at interacting with the real world. These are usually rigid or made of metal. The team wanted to explore the space of using compliant, rubber-like materials.

During test dives in Rainbow Reef in Fiji, SoFi swam at depths of more than 15m (50-feet) for up to 40 minutes at once, easily handling currents and taking high-resolution photos and videos using a fish-eye lens. Using its undulating tail and a unique ability to control its own buoyancy, SoFi could swim in a straight line, turn, or dive up or down. The researchers also used a waterproofed Super Nintendo controller and developed a custom acoustic communications system that enabled them to change SoFi's speed and have it make specific moves and turns.

Katzschmann worked on the project and wrote the paper with MIT's Computer Science and Artificial Intelligence Laboratory director Daniela Rus, graduate student Joseph DelPreto and former post-doc Robert MacCurdy, who is now assistant professor at University of Colorado at Boulder.

Lunar, the AR-enabled moon model

AstroReality has launched the world's first smart moon model featuring a high-resolution topography. Each topographic feature is perfected by hand before it is finished with hand-applied colours and a safe, protective coating.



Through AstroReality app and its AR function, Lunar provides you the sight, sound and touch of the moon (Credit: www.indiegogo.com)

Lunar models are 1:116,020,845 in scale, high-definition, 3D-printed and hand-crafted. Using the most accurate industrial 3D printing technology, Lunar is modelled to 0.006 millimetres per pixel precision, based on data captured by NASA's Lunar Reconnaissance Orbiter.

Using advanced augmented reality (AR) technology, Lunar, paired with AstroReality app, weaves together an unparalleled way to experience the moon, in your hands, and through your smartphone. It allows you to learn captivating facts and trivia, simulate moon missions and embark on personalised lunar adventures from your home.